

PROG2510 – WINDOWS DESKTOP PROGRAMMING

PROJECT

Groups of 2 or 3

OVERVIEW

Messaging systems are common tools for communication in today's high-tech world. In most cases, these systems are Internet-based.

In this project, you will be developing a LAN-based messaging system to demonstrate several of the concepts presented in this course.

SUPPORTING COURSE OUTCOMES

- Create Windows applications using the .NET Framework.
- Implement simple threaded solutions in Windows applications.
- Compare Windows Services with Windows desktop applications from a development perspective.

ACADEMIC INTEGRITY AND LATE PENALTIES

- Please refer to the **What is Academic Integrity** section in the [Applied Computer Science & Information Technology Academic Handbook](#)
- Please refer to the **Assignment Submission Standards** section in the [Applied Computer Science & Information Technology Academic Handbook](#)

USE OF GENERATIVE AI

- You are prohibited from using generative AI tools to create any part of the solution to this assignment.
- You are permitted to use generative AI tools for personal use to learn skills and gain knowledge related to the course content
- Please refer to the Generative AI Toolkit for Students ([Home - Gen AI for Students - Library & Learning Services at Conestoga College](#))

EVALUATION

- Project is worth 30% of the final mark
- For details on evaluating the project, please see the rubric in eConestoga

PREPARATION

- Review related concepts covered in the course

ASSIGNMENT REQUIREMENTS

The *service* must be created using the .NET Framework (please use at least version 4.72). There is no template for .NET (Core) for Services.

The *client* component must be based on .NET (Core) version 8.0 or later (*not .NET Framework*).

You must follow SET coding standards, especially with respect to comments.

You cannot use **var** to declare any variables.

You cannot have more than one **return()** statement in a method, and it must be the last statement in the method.

If you use ANY code segments from a source other than the course material, you **MUST** cite it.

You must follow SET coding standards, especially with respect to comments.

Here are the specific requirements:

1. The solution will present a simple text messaging system for a Windows-based system. The general approach is as follows:
 - a. Communication protocol is TCP/IP. You **MUST** implement the protocol using **synchronous** methods.
 - b. The server is implemented as a service. It manages the flow of messages between clients.
 - c. A minimum of two clients must be demonstrated working, with the users sending messages to each other.
 - d. The messaging protocol must be a disconnected model, following a request-response approach.
2. The server...
 - a. Must be implemented as a service.
 - b. Must be able to get the IP address and port the configuration file.
 - c. Receives a message from a client and responds with an acknowledgement.
 - d. Sends a message to its intended target and waits for an acknowledgement.
 - e. Makes use of an event log, writing to a text file specified in the configuration file. The log entry must include (at a minimum) the date and time the log was created, and a message as passed to the log write method.
3. The client...
 - a. Must be a WPF application. As usual, it must resemble a typical Windows application.
 - b. The server IP address and Port must be managed through settings in the client application. You may choose how you will implement this.
 - c. The User Interface must include a Menu (with a Help->About menu option implemented), a control that shows all messages between the user and another

user, and a control that allows the user to enter a message to be sent to another user.

- d. The message entered by the user is sent to the server.
- e. A message sent by another user is received from the server

SUBMISSION REQUIREMENTS

- There are no specific file naming requirements for the submission file. Please keep the name short, but meaningful.
- Create two projects – one with the server service and the other with the client software. You may put them in one solution or two separate solutions.
- Please zip the solution(s) into one zip folder for submission